

Continuous Positive Airway Pressure (CPAP) Device

ACG: A-0431 (AC)

[Link to Codes](#)

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Clinical Indications

- Continuous positive airway pressure (CPAP) may be indicated for **1 or more** of the following(1)(2):
 - ☐ Central sleep apnea, and CPAP needed, as indicated by **1 or more** of the following(11)(12)(13):
 - Primary (ie, idiopathic) central sleep apnea and **ALL** of the following:
 - Primary (ie, idiopathic) central sleep apnea on diagnostic polysomnography, as indicated by **ALL** of the following(13):
 - Central apnea-hypopnea index of 5 or greater(14)(15)(16)
 - Central apneas plus central hypopneas represent 50% or more of all apneas and hypopneas.
 - No Cheyne-Stokes respiration[A]
 - No nocturnal hypoventilation (ie, no hypercapnia),[B] as indicated by **1 or more** of the following(20):
 - Adult does not have sleep PCO₂ greater than 55 mm Hg (7.3 kPa) for 10 minutes or longer, or sleep PCO₂ increasing by at least 10 mm Hg (1.3 kPa) (vs awake supine PCO₂) and above 50 mm Hg (6.7 kPa) for 10 minutes or longer.
 - Child does not have PCO₂ greater than 50 mm Hg (6.7 kPa) for more than 25% of total sleep time.
 - Symptoms of sleep-disordered breathing, as indicated by **1 or more** of the following:
 - Daytime sleepiness
 - Frequent nighttime awakenings
 - Gasping during sleep
 - Insomnia
 - Respiratory pauses during sleep
 - Restless sleep
 - Titration study with polysomnography demonstrates improvement in apnea-hypopnea index while using CPAP. (21)
 - No evidence of medical disorders or medications that explain symptoms(22)
 - Secondary central sleep apnea due to heart failure, as indicated by **ALL** of the following(23)(24):
 - CPAP shown to be effective during sleep laboratory CPAP titration study, including reduction of apnea-hypopnea index below 15 per hour(22)

- Inadequate response to aggressive medical management(25)
- ☐ Obesity hypoventilation syndrome, and CPAP needed, as indicated by **1 or more** of the following(3)(26)(27)(28)(29):^[N]
 - Initial therapy,^{[C][D]} as indicated by **ALL** of the following:
 - BMI greater than 30  BMI Calculator
 - Daytime hypercapnia with PaCO₂ greater than 45 mm Hg (6.0 kPa) without other etiology (eg, kyphoscoliosis, lung parenchymal disease, myopathy, severe hypothyroidism)
 - Severe obstructive sleep apnea component on polysomnography (ie, apnea-hypopnea index of 30 or greater) (31)(37)
 - TSH level does not demonstrate hypothyroidism.
 - Subsequent therapy,^[E] with annual provider assessment of adherence and effectiveness performed (eg, symptom resolution, tolerance of expiratory pressure, device usage) and potential barriers to adherence or effectiveness addressed(7)(36)(40)(45)
- ☐ Obstructive sleep apnea, and CPAP needed, as indicated by **1 or more** of the following(46)(47):^[I]
 - Initial therapy,^[C] as indicated by **1 or more** of the following:
 - Adult and **1 or more** of the following(43)(63):
 - Obstructive sleep apnea present, as indicated by **1 or more** of the following(64):
 - Mild obstructive sleep apnea (ie, apnea-hypopnea index or respiratory disturbance index between 5 and less than 15, determined with polysomnography using adult scoring rules)^[F] and **1 or more** of the following(20)(66):
 - Cardiovascular disease has been documented (eg, hypertension, ischemic heart disease, heart failure, stroke).(67)
 - Excessive daytime sleepiness^[G](68)(69)(70)
 - Fibromyalgia-like symptoms(71)
 - Headaches upon awakening
 - Heartburn and reflux
 - Impaired cognition(72)
 - Mood disorder
 - Night sweats
 - Nocturia or nocturnal enuresis(73)
 - Observed apnea or choking episodes
 - Patient is commercial vehicle driver.(74)
 - Snoring
 - Moderate or severe obstructive sleep apnea (ie, apnea-hypopnea index or respiratory disturbance index 15 or greater, determined with polysomnography using adult scoring rules) ^[F](20)(75)(76)
 - Upper airway resistance syndrome and unexplained excessive daytime sleepiness(77)(78)(79)(80)
 - Child or adolescent^{[H][I]} and **ALL** of the following(57)(88)(91):
 - Age younger than 18 years
 - Obstructive sleep apnea present, as indicated by **1 or more** of the following:
 - Mild obstructive sleep apnea (ie, apnea-hypopnea index from 1 to 5) and **1 or more** of the following(87):
 - Achondroplasia
 - Behavioral problems
 - Cardiovascular disease (eg, elevated blood pressure, pulmonary hypertension)
 - Chiari malformation
 - Craniofacial abnormalities
 - Down syndrome
 - Excessive daytime sleepiness
 - Impaired cognition
 - Inattention or hyperactivity
 - Mucopolysaccharidoses
 - Neuromuscular disorders
 - Prader-Willi syndrome
 - Moderate or severe obstructive sleep apnea (ie, apnea-hypopnea index greater than 5)(56)(57)(85)(92)
 - Residual apnea-hypopnea index greater than 5 in pediatric patient after adenotonsillectomy(93)(94)(95)(96)(97)
 - Subsequent therapy,^[E] with annual provider assessment of adherence and effectiveness performed (eg, symptom resolution, tolerance of expiratory pressure, device usage) and potential barriers to adherence or effectiveness addressed(7)(36)(39)(40)(45)(98)

Alternatives

- Alternatives include(99)(100)(101):
 - For adult with mild obstructive sleep apnea: oral appliance.(49)(102)(103) See Oral Appliances (Mandibular Advancement Devices) [↗](#) AC for further information.
 - For adult with mild, moderate, or severe obstructive sleep apnea: nasal expiratory resistance device or BPAP device. See Nasal Expiratory Resistance Device [↗](#) AC and Bilevel Positive Airway Pressure (BPAP) Device [↗](#) AC for further information.
 - For adult with mild, moderate, or severe obstructive sleep apnea and tonsils or adenoids contributing: adenoidectomy or tonsillectomy. See Adenoidectomy [↗](#) AC or Tonsillectomy [↗](#) AC for further information.
 - For child with obstructive sleep apnea: adenoidectomy and tonsillectomy, or BPAP device.(92)(104) See Adenoidectomy [↗](#) AC, Tonsillectomy [↗](#) AC, and Bilevel Positive Airway Pressure (BPAP) Device [↗](#) AC for further information.
 - For infant with obstructive sleep apnea (ie, craniofacial abnormalities): adenoidectomy, mandibular distraction, supraglottoplasty, or tracheostomy.(61)(92)(105) See Adenoidectomy [↗](#) AC and Mandibular Osteotomy [↗](#) AC for further information.
 - For obesity hypoventilation syndrome: BPAP device, tracheostomy, or weight loss.(28)(49) See Bilevel Positive Airway Pressure (BPAP) Device [↗](#) AC for further information.
 - For obstructive sleep apnea, and inadequate response to or intolerance of CPAP, or concomitant sleep hypoventilation: BPAP device. See Bilevel Positive Airway Pressure (BPAP) Device [↗](#) AC for further information.

Evidence Summary

Background

Continuous positive airway pressure (CPAP) delivers a fixed level of pressure to the airway throughout inspiration and expiration; this serves to stent a collapsible airway while also positively impacting atelectasis, improving oxygenation, decreasing the work of breathing, and reducing left ventricular afterload. It also increases residual capacity, reducing ventilation-perfusion mismatch.(3)(4)(5) **(EG 2)** A variety of interfaces may be used to connect the patient to the device at home (eg, combined mouth-nasal cradle mask, nasal mask, nasal pillows). Identifying the best interface for the patient during the CPAP titration phase may improve adherence; problems with adherence may require a change in interface or a retitration study.(6)(7)(8)(9) **(EG 2)** A change in interface style may be required if a pressure injury occurs (eg, skin breakdown, skin irritation, dental pain). Leaks that develop over time may indicate cushion deterioration and the need for replacement.(6)(10) **(EG 2)** CPAP is administered as a first-line therapy to treat obstructive sleep apnea.(4) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 74 published peer reviewed articles, 12 specialty society or other evidence-based guidelines, 1 Cochrane systematic review, and 9 book sections.

For central sleep apnea, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review identified 16 studies evaluating the effect of CPAP on central sleep apnea syndromes related to heart failure and concluded that there was moderate evidence for the use of CPAP therapy targeted to normalize the apnea-hypopnea index, with documented improvement in left ventricular ejection fraction and a positive effect on transplant-free survival.(11) **(EG 1)**

For obesity hypoventilation syndrome, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and meta-analysis identified 3 randomized controlled trials that compared BPAP with CPAP in 247 patients with obesity hypoventilation syndrome. CPAP provided similar improvement in daytime PaCO₂, Epworth Sleepiness Scale scores, Functional Outcomes of Sleep Questionnaire scores, and total sleep time spent with oxygen saturation below 90% as compared with BPAP. The authors noted that larger, longer-term studies were needed to confirm the findings.(30) **(EG 1)** A multicenter, open-label, randomized controlled trial compared CPAP with volume-targeted pressure support ventilation in 215 patients with obesity hypoventilation syndrome complicated by an apnea-hypopnea index of 30 or greater and reported, at a median follow-up of 5.44 years, that mean hospital days were similar between the 2 arms. Blood pressure, Epworth Sleepiness Scale scores, hypercarbia, and quality of life improved in both arms without differences between the 2 arms (median follow-up 3 years). Therapy complexity was lower in the CPAP arm.(31) **(EG 1)** A network meta-analysis of 7 randomized controlled trials in adults with obesity hypoventilation syndrome evaluating different positive airway pressure devices (ie, BPAP, BPAP with backup rate, BPAP with volume-assured pressure support, CPAP) reported that daytime hypercapnia and Epworth Sleepiness Scale scores showed statistically greater improvement with the use of BPAP and BPAP with volume-assured pressure support; however, the authors noted that the differences were small and the clinical significance of the findings was not established.(32) **(EG 1)** A specialty society guideline noted that significant weight loss may allow for a transition to a different device (eg, BPAP, CPAP) with polysomnographic guidance.(31) **(EG 1)** Patients with obesity hypoventilation syndrome who have a mild obstructive sleep apnea component on sleep study may respond to first-line noninvasive ventilation. First-line CPAP therapy is often used for patients who have a greater number of obstructive events on sleep study.(33)(34)(35) **(EG 2)** An evidence-based specialty society guideline on obesity hypoventilation syndrome recommended the use of first-line CPAP in stable obesity hypoventilation syndrome patients with a severe obstructive sleep apnea component (ie, apnea-hypopnea index of 30 or

greater); it also noted that the most appropriate first-line positive airway pressure therapy for patients with a less severe obstructive sleep apnea component has not been established.(28) **(EG 2)**

For obstructive sleep apnea, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Nocturnal CPAP is a standard of care, with review articles, randomized studies, and guidelines confirming its ability to improve sleep-disordered breathing, abnormal oxygenation, fatigue, sleepiness, and other important clinical outcome parameters.(47)(48)(49)(50) **(EG 1)** It is also indicated to maintain a patent oropharynx in obstructive sleep apnea or hypoventilatory syndromes.(40)(47)(51) **(EG 2)** A systematic review found no evidence of improved long-term health outcomes (eg, all-cause mortality, cardiovascular mortality, cognitive function, composite cardiovascular outcomes, incident diabetes, myocardial infarction risk, stroke risk) with use of CPAP based on heterogeneous underpowered data from randomized controlled trials. Also noteworthy was the lack of evidence available to demonstrate that improvement in apnea-hypopnea index was a valid surrogate for long-term health outcomes or to understand how obstructive sleep apnea severity and comorbidity severity impact CPAP effectiveness and might inform patient selection. The authors recommended that further high-quality trials be performed to clarify these issues.(52) **(EG 1)** A systematic review and meta-analysis identified 25 randomized trials (1596 patients) that compared CPAP with conventional management of obstructive sleep apnea and reported significant improvement in apnea-hypopnea events with the use of CPAP, most noticeably in patients with moderate or severe obstructive sleep apnea; 13 randomized trials (746 patients) compared CPAP with mandibular advancement in patients with moderate to severe obstructive sleep apnea and reported a significantly lower apnea-hypopnea index in the CPAP arm.(53) **(EG 1)** A multicenter randomized controlled trial including 233 adults with mild obstructive sleep apnea (ie, apnea-hypopnea index from 5 to 15) compared sleep counseling with or without 3 months of CPAP therapy and found that patients in the CPAP arm reported significant improvements in quality of life (as measured by the 36-Item Short Form Health Survey) compared with controls.(54) **(EG 1)** A joint society guideline notes that untreated obstructive sleep apnea may double the risk for stroke; however, recent randomized controlled trials have not demonstrated stroke risk reduction with the use of CPAP. Evidence suggests that subgroups with good CPAP adherence may benefit from CPAP therapy.(55) **(EG 2)** A review article noted that CPAP has been shown to be as effective as adenotonsillectomy in reducing the apnea-hypopnea index in children with obstructive sleep apnea; however, adherence in children is often problematic.(56) **(EG 2)** A specialty society task force on sleep-disordered breathing in childhood recommended the use of CPAP in children who have a residual apnea-hypopnea index greater than 5 events per hour after tonsillectomy and in children with obstructive sleep apnea secondary to craniofacial abnormalities or neuromuscular disorders; however, BPAP is indicated if hypercapnia is present during sleep, indicating nocturnal hypoventilation.(57) **(EG 2)** Review articles and a specialty society guideline note that patients who are intolerant of the pressure or unresponsive to CPAP (ie, ongoing obstructive events on polysomnography despite the use of 15 cm H₂O (1471 Pa)) may be appropriate for a trial of BPAP; BPAP uses lower expiratory pressure against which the patient must exhale.(47)(58)(59) **(EG 2)** A review article notes that less than 10% of patients with obstructive sleep apnea will require a more aggressive respiratory support device than CPAP, such as BPAP.(60) **(EG 2)** CPAP may not be feasible for managing obstructive sleep apnea in some infants due to sleep-wake patterns in infants or limited interface sizes available for infants; however, the availability of infant-sized masks has improved.(61)(62) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(106); 42 CFR 422.101(107)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(108); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(109); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(110)

Medicare Coverage Determinations: Medicare Coverage Database(111)

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Footnotes

[A] Cheyne-Stokes respiration is characterized by crescendo-decrescendo tidal volume breathing with interposed central apneas or hypopneas. As a result of the decrescendo breathing, hypopneas, and apneas, PCO₂ rises and PO₂ falls, which stimulates chemoreceptors, in turn producing the crescendo breathing pattern.(17) [A in Context Link 1]

[B] Patients with primary central sleep apnea are hypocapnic and do not develop sleep hypercapnia.(18) Polysomnography in primary central sleep apnea typically reveals central apneas in isolation or in clusters; no daytime or sleep hypoventilation is observed.(19) [B in Context Link 1]

[C] Assessing clinical outcomes early in the course of treatment with CPAP may allow a clinician to evaluate the patient's response to CPAP and address potential barriers to adherence.(36) [C in Context Link 1, 2]

[D] For patients with obesity hypoventilation syndrome, improvements in blood gases are expected within 3 months of initiation of CPAP therapy.(33) [D in Context Link 1]

[E] Follow-up of patients includes clinician visits and physiologic monitoring as determined by the underlying condition, patient status, and clinical features. Re-evaluation of the patient is performed at least annually and whenever concerns arise.(7)(38)(39)(40)(41)(42)(43) In some conditions, patients may improve sufficiently on follow-up evaluation to discontinue CPAP support altogether (eg, after weight loss in patients with obesity, child with spontaneous improvement over time).(38)(39)(40)(44) [E in Context Link 1, 2]

[F] The apnea-hypopnea index is defined as the number of obstructive apneas and hypopneas per hour of recorded sleep, which is recorded during polysomnography. The respiratory disturbance index is the total number of breathing disturbances per hour of monitoring time (ie, the number of apneas, hypopneas, and respiratory event-related arousals, the last being characterized by increased respiratory efforts leading to arousals from sleep that do not fulfill the criteria for apnea or hypopnea). In adults, mild obstructive sleep apnea is defined as apnea-hypopnea index or respiratory disturbance index of 5 to less than 15, moderate as 15 to less than 30, and severe as 30 or greater.(47)(65) [F in Context Link 1, 2]

[G] The Epworth Sleepiness Scale grades the chances of falling asleep (from 0 to 3) during 8 sedentary activities, such as watching television.(65) [G in Context Link 1]

[H] A specialty society recommends using the pediatric scoring rules to categorize sleep events (eg, apnea, hypopnea) for children younger than 18 years; however, studies indicate that some of the adult scoring rules may be used in adolescents 13 to 18 years of age.(20)(81)(82)(83)(84) After scoring is completed, pediatric polysomnogram cutoffs are used to interpret sleep studies in patients younger than 18 years (ie, mild obstructive sleep apnea is defined as an apnea-hypopnea index of 1 to 5, moderate as 6 to 10, and severe as greater than 10).(57)(85)(86) The need for research into outcomes of the pediatric clinical classification and sleep events has been highlighted; the current classification is based on consensus.(87)(88) [H in Context Link 1]

[I] Stand-alone CPAP devices have minimum body weight constraints, which may preclude their use in younger patients. Multimodal home ventilators that include a CPAP mode may be appropriate in these patients.(89) A specialty society guideline notes that, when prescribed by pediatric sleep experts, CPAP devices have been reported to be safe and effective for pediatric patients whose age and/or body weight is considered to be off-label for the device. These devices may include software algorithms that estimate adherence and efficacy (eg, improvement of apnea-hypopnea index); however, in pediatric patients, these algorithms may be inaccurate.(90) [I in Context Link 1]

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